

WENRUI XU

Minneapolis, Minnesota

☎ (651) 206-1990

✉ xu000424@gmail.com

🌐 [linkedin.com/in/wenrui-xu-6656451a1](https://www.linkedin.com/in/wenrui-xu-6656451a1)

🎓 [Google Scholar](#)

Education

University of Minnesota, Twin Cities

Ph.D. in Electrical Engineering, GPA: 3.9/4.0

- **Concentration:** Machine Learning, Time-Series Modeling, and Statistical Signal Processing
- **Advisor:** Prof. Keshab K. Parhi

Expected 2028

Minneapolis, Minnesota

University of Minnesota, Twin Cities

B.S. in Computer Engineering, GPA: 3.9/4.0

May 2023

Minneapolis, Minnesota

Industry Experience

Boston Scientific

Rhythm Management R&D Research Intern

Summers 2023 – 2025

Arden Hills, Minnesota

Longitudinal Time-Series Prediction

May 2025 – August 2025

- Developed deep sequence models for forecasting and classification from multi-year, irregular time-series data.
- Analyzed long-horizon dependencies, cross-sectional heterogeneity, and out-of-sample generalization across unseen populations.
- Compared model architectures through controlled experiments, cross-validation, threshold analysis, and subgroup evaluation.
- Built scalable PyTorch pipelines for preprocessing, model training, hyperparameter search, and reproducible performance analysis.

Automated Research and Information Extraction

May 2024 – August 2024

- Built an LLM-based research application integrating external APIs, automated document processing, and an interactive interface.
- Developed structured extraction and summarization workflows to identify methods and quantitative results from technical literature.
- Accelerated exploratory research by automating document ingestion, comparison, and synthesis across large collections of publications.

Large-Scale Data Processing and Predictive Modeling

May 2023 – August 2023

- Developed convolutional models for multiclass prediction from high-dimensional, multichannel time-series observations.
- Designed a parallel preprocessing pipeline for a 2.5 TB database, achieving a 40× speedup and enabling large-scale experimentation.
- Implemented reproducible TensorFlow workflows for cleaning, normalization, feature extraction, training, and model evaluation.

Research Experiences

Robust Time-Series Representation Learning

2025 – Present

- Developed an encoder-agnostic diffusion framework for learning stable representations from noisy and nonstationary sequential data.
- Formulated consistency objectives that minimize representation drift under perturbations and distribution shift.
- Designed prototype-based and subspace-constrained objectives to improve out-of-sample and cross-subject generalization.
- Implemented PyTorch pipelines for cross-validation, hyperparameter optimization, experiment tracking, and systematic ablation studies.

Hyperdimensional Learning for Large-Scale Time-series Data

2022 – 2024

- Designed hyperdimensional computing algorithms for classification and anomaly detection across eight benchmark datasets.
- Combined temporal, frequency-domain, wavelet, and statistical features with supervised feature-selection methods.
- Improved anomaly-detection accuracy by up to 28%, F1 score by up to 23%, and AUC by up to 10% over prior methods.
- Reduced training time by 3–4× and memory consumption by up to 10× through compact representations and efficient computation.
- Conducted controlled comparisons across feature subsets, encoding strategies, and representation dimensions.

Graph Representation Learning and Link Prediction

2023 – 2024

- Implemented hyperdimensional representations for link prediction on FB15k-237 and WN18RR datasets.
- Compared encoding methods, similarity functions, and representation dimensions for scalable knowledge-graph link prediction.
- Analyzed computational efficiency, scalability, and model limitations relative to established graph-learning baselines.

Publications

- **W. Xu** and K. K. Parhi, “A Survey of Attacks on Large Language Models,” *arXiv preprint arXiv:2505.12567*, 2025
- A. Chatterjee, N. K. Jha, J. D. Cohen, T. L. Griffiths, H. Lu, D. Marculescu, A. Rasul, **W. Xu**, and K. K. Parhi, “A Vision Toward Energy-Efficient Domain-Specific Artificial Intelligence Models and Agents,” *arXiv preprint arXiv:2510.22052*, 2026
- **W. Xu**, E. Krainess, A. Payani, H. Latapie, and K. K. Parhi, “IFODHD: Improved Feature Selection Based Outlier Detection using Hyperdimensional Computing,” *Journal of Signal Processing Systems*, vol. 96, no. 12, pp. 791–807, 2024
- **W. Xu** and K. K. Parhi, “Multi-Class Classification of Abnormal Heartbeat Detection using Hyperdimensional Computing,” *Journal of Signal Processing Systems*, vol. 96, no. 8, pp. 463–477, 2024

Technical Skills

Programming: Python, C/C++, MATLAB, SQL, Java

AI/ML: PyTorch, TensorFlow, Time-Series Modeling, Diffusion Models, State-Space Models, Representation Learning, Hyperdimensional Computing

Engineering & Systems: Signal Processing, Parallel Computing, Linux, Git, Docker